pH Electrodes

Body and Material Variations

Glass Bodies: Standard in laboratories for high accuracy and wide chemical/temperature compatibility, though they are fragile.

Epoxy/Polymer Bodies: Built with rugged plastic housings to resist breakage in industrial process control, field work, or rugged environments.

A pH meter is an instrument used to measure the acidity or alkalinity (basicity) of a solution, expressed as pH.
The pH scale ranges from 0 to 14: pH
< 7: Acidic
pH = 7: Neutral (e.g., pure water)
pH > 7: Alkaline (basic)
Main Components of a pH Meter:
1. Glass Electrode: The sensor placed in the solution to measure the voltage related to hydrogen ion concentration.
2. Reference Electrode: Provides a stable voltage for comparison.
3. Display Unit: Shows the pH reading.
4. Calibration Controls: Used to calibrate the meter with standard buffer solutions (typically pH 4, 7, and 10).
How to Use a pH Meter:
1. Calibrate the meter using standard buffer solutions.
2. Rinse the electrode with distilled water before and after each measurement.
3. Place the electrode into the sample solution.
4. Wait for the reading to stabilize.
5. Read the pH on the display.
Important Tips:
Store the electrode in a proper storage solution, not in distilled water.
Clean the electrode regularly to ensure accuracy.
Regular calibration is necessary for precise results.
Examples where pH measurements taken.
Water and waste water treatment
Petroleum laboratories
Chemical manufacturing
Food and beverage testing

The most commonly used electrode for an electrocardiogram (ECG) is the disposable silver/silver chloride (Ag/AgCl) electrode.

